The documentation and process conversion measures necessary to comply with this revision shall be completed by 9 April 2007.

INCH-POUND

MIL-PRF-19500/476D
9 January 2007
SUPERSEDING
MIL-PRF-19500/476C
9 April 2003

* PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, P-CHANNEL, SILICON, TYPES 2N5114 THROUGH 2N5116 AND 2N5114UB THROUGH 2N5116UB, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for P-channel, junction, silicon field-effect transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-18) and figure 2 (UB, surface mount).

1.3 Maximum ratings. $T_A$ = +25°C, unless otherwise specified.

| $P_T$ (1)<br>$T_A$ = +25°C | $V_{GS}$ (2) | $V_{DS}$ | $V_{DG}$ (2) | $I_G$ | $T_{STG}$ |
|---|---|---|---|---|---|
| W | V dc | V dc | V dc | mA dc | °C |
| 0.500 | 30 | 30 | 30 | 50 | -65 to +200 |

(1) Derate linearly 3.0 mW/°C for $T_A$ = +25°C.
(2) Symmetrical geometry allows operation of those units with source/drain leads interchanged.

1.4 Primary electrical characteristics.

| Limits | $I_{DSS}$ (1) | | | $V_{DS(on)}$ | | |
|---|---|---|---|---|---|---|
| | $V_{DS}$ = -18 V dc<br>$V_{GS}$ = 0 | $V_{DS}$ = -15 V dc<br>$V_{GS}$ = 0 | $V_{DS}$ = -15 V dc<br>$V_{GS}$ = 0 | $I_D$ = -15 mA dc<br>$V_{GS}$ = 0 | $I_D$ = -7 mA dc<br>$V_{GS}$ = 0 | $I_D$ = -3 mA dc<br>$V_{GS}$ = 0 |
| | 2N5114<br>2N5114UB | 2N5115<br>2N5115UB | 2N5116<br>2N5116UB | 2N5114<br>2N5114UB | 2N5115<br>2N5115UB | 2N5116<br>2N5116UB |
| | mA dc | mA dc | mA dc | V dc | V dc | V dc |
| Min | -30 | -15 | -5.0 | | | |
| Max | -90 | -60 | -25 | -1.3 | -0.8 | -0.6 |

See notes on next page.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.



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1.4 Primary electrical characteristics - Continued.

| Limits | $V_{GS(off)}$ | | | $r_{ds(on)2}$ | | |
|---|---|---|---|---|---|---|
| | $V_{DS}$ = 15 V dc<br>$I_D$ = -1.0 nA dc | | $V_{DS}$ = -15 V dc<br>$I_D$ = 1.0 µA dc | $V_{GS}$ = 0, $I_D$ = 0<br>f = 1 kHz | | |
| | 2N5114<br>2N5114UB | 2N5115<br>2N5115UB | 2N5116<br>2N5116UB | 2N5114<br>2N5114UB | 2N5115<br>2N5115UB | 2N5116<br>2N5116UB |
| | V dc | V dc | V dc | Ω | Ω | Ω |
| Min | 5 | 3 | 1 | | | |
| Max | 10 | 6 | 4 | 75 | 100 | 175 |

(1) For $T_A$ > +25°C, derate linearly 1.67 mW/°C one section, 2.67 mW/°C both sections.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

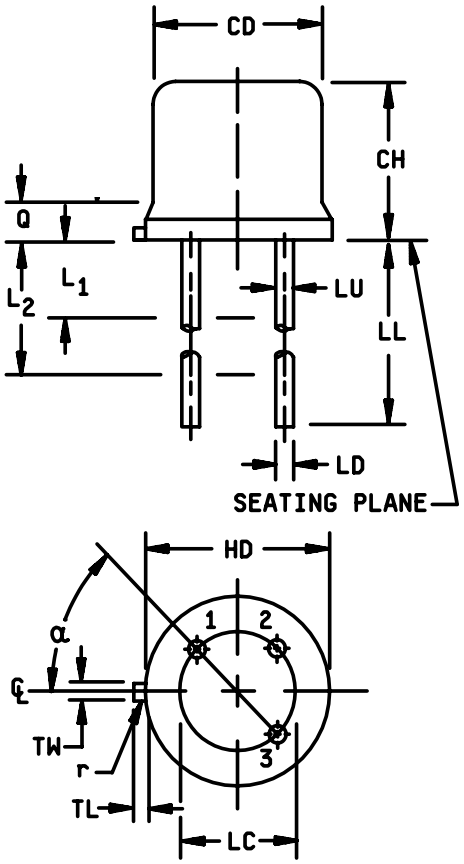
* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7,8 |
| LL | .500 | .750 | 12.7 | 19.05 | 7,8 |
| LU | .016 | .019 | 0.41 | 0.48 | 7,8 |
| $L_1$ | | .050 | | 1.27 | 7,8 |
| $L_2$ | .250 | | 6.35 | | 7,8 |
| Q | | .030 | | 0.76 | 5 |
| TL | .028 | .048 | 0.71 | 1.22 | 3,4 |
| TW | .036 | .046 | 0.91 | 1.17 | |
| r | | .010 | | 0.25 | 10 |
| $\alpha$ | 45° TP | | 45° TP | | 6 |

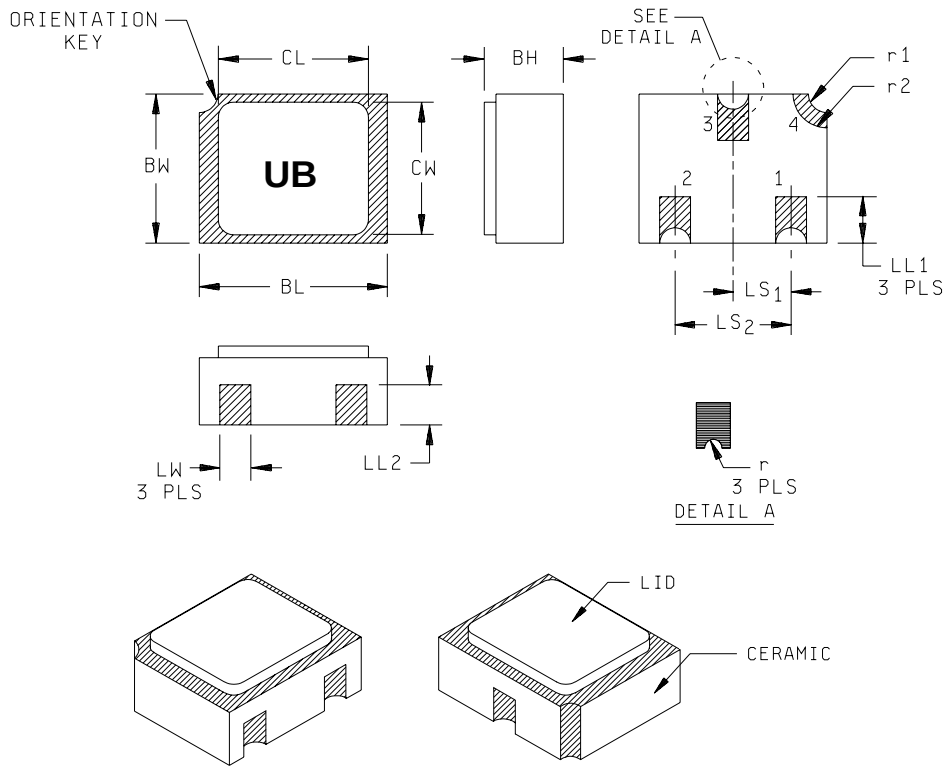



NOTES:
1. Dimension are in inches.
* 2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown on figure 2.
7. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. The gate shall be electrically connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.
12. Lead 1 = source, lead 2 = gate, lead 3 = drain.

* FIGURE 1. Physical dimensions (similar to TO-18).



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | |
| BL | .115 | .128 | 2.92 | 3.25 | |
| BW | .085 | .108 | 2.16 | 2.74 | |
| CL | | .128 | | 3.25 | |
| CW | | .108 | | 2.74 | |
| LL1 | .022 | .038 | 0.56 | 0.96 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | |
| LS1 | .036 | .040 | 0.91 | 1.02 | |
| LS2 | .071 | .079 | 1.81 | 2.01 | |
| LW | .016 | .024 | 0.41 | 0.61 | |
| r | | .008 | | .203 | |
| r1 | | .012 | | .305 | |
| r2 | | .022 | | .559 | |

NOTES:
1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas (tungsten with gold plating 60 micro inches min over 80 micro inches min nickel.
* 4. Pad 1 = drain, Pad 2 = source, Pad 3 = gate, Pad 4 = shielding connected to the lid.

* FIGURE 2. Physical dimensions, surface mount (UB version).



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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (similar to TO-18) and 2 (UB, surface mount) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, and JANTXV can be abbreviated as J, JX, and JV respectively. The "2N" prefix and the "UB" suffix can also be omitted.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500 | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| 9 and 10 | Not applicable |
| 11 | $I_{D(off)1}$, $r_{ds(on)1}$ |
| 12 | See 4.3.1 |
| 13 | Subgroup 2 of table I herein;<br>$\Delta I_{D(off)1}$ = ±0.5 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{ds(on)1}$ = ±20 percent of initial value. |

4.3.1 Power burn-in. Power burn-in conditions are in accordance with method 1039 of MIL-STD-750, condition A and as follows: $T_A$ = +150°C; $V_{GS}$ = 24 V dc; $V_{DS}$ = 0.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.



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4.4.2.1 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | $V_{GS}$ = 24 V dc; $T_A$ = +125°C; $V_{DS}$ = 0. |
| B6 | 1032 | $T_A$ = +175°C. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| Subgroup | Method | Condition |
|---|---|---|
| * C2 | 1056 | Thermal shock: Test condition C, time at temperature extremes = 15 minutes minimum. |
| C2 | 2036 | Lead fatigue: Test condition E (not applicable to UB suffix devices.) |
| C6 | 1026 | $V_{GS}$ = 24 V dc; $T_A$ = +125°C; $V_{DS}$ = 0. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Gate to source breakdown voltage | 3401 | Bias condition C; $I_G$ = 1.0 μA dc; $V_{DS}$ = 0 | $V_{(BR)GSS}$ | 30 | | V dc |
| Gate reverse current | 3411 | Bias condition C; $V_{GS}$ = 20 V dc; $V_{DS}$ = 0 | $I_{GSS}$ | | 500 | pA dc |
| Drain cutoff current | 3413 | Bias condition A | $I_{D(off)1}$ | | | |
| 2N5114, 2N5114UB | | $V_{DS}$ = -15 V dc; $V_{GS}$ = 12 V dc | | | -500 | pA dc |
| 2N5115, 2N5115UB | | $V_{DS}$ = -15 V dc; $V_{GS}$ = 7.0 V dc | | | -500 | pA dc |
| 2N5116, 2N5116UB | | $V_{DS}$ = -15 V dc; $V_{GS}$ = 5.0 V dc | | | -500 | pA dc |
| Zero-gate-voltage drain current | 3413 | Bias condition C; $V_{DS}$ to be pulsed see 4.5.1 | $I_{DSS}$ | | | |
| 2N5114, 2N5114UB | | $V_{DS}$ = -18 V dc; $V_{GS}$ = 0 | | -30 | -90 | mA dc |
| 2N5115, 2N5115UB | | $V_{DS}$ = -15 V dc; $V_{GS}$ = 0 | | -15 | -60 | mA dc |
| 2N5116, 2N5116UB | | $V_{DS}$ = -15 V dc; $V_{GS}$ = 0 | | -5.0 | -25 | mA dc |
| Drain to source on voltage | 3405 | | $V_{DS(on)}$ | | | |
| 2N5114, 2N5114UB | | $I_D$ = -15 mA dc; $V_{GS}$ = 0 | | | -1.3 | V dc |
| 2N5115, 2N5115UB | | $I_D$ = -7.0 mA dc; $V_{GS}$ = 0 | | | -0.8 | V dc |
| 2N5116, 2N5116UB | | $I_D$ = -3.0 mA dc; $V_{GS}$ = 0 | | | -0.6 | V dc |
| Gate to source cutoff voltage | 3403 | | $V_{GS(off)}$ | | | |
| 2N5114, 2N5114UB | | $V_{DS}$ = -15 V dc; $I_D$ = -1.0 nA dc | | 5.0 | 10 | V dc |
| 2N5115, 2N5115UB | | $V_{DS}$ = -15 V dc; $I_D$ = -1.0 nA dc | | 3.0 | 6.0 | V dc |
| 2N5116, 2N5116UB | | $V_{DS}$ = -15 V dc; $I_D$ = -1.0 nA dc | | 1.0 | 4.0 | V dc |
| Small-signal drain to source on resistance | 3423 | $V_{GS}$ = 0; $I_D$ = -1 mA dc | $r_{ds(on)1}$ | | | |
| 2N5114, 2N5114UB | | | | | 75 | Ω |
| 2N5115, 2N5115UB | | | | | 100 | Ω |
| 2N5116, 2N5116UB | | | | | 175 | Ω |
| Small-signal drain to source on resistance | 3423 | $V_{GS}$ = 0; $I_D$ = 0; f = 1 kHz | $r_{ds(on)2}$ | | | |
| 2N5114, 2N5114UB | | | | | 75 | Ω |
| 2N5115, 2N5115UB | | | | | 100 | Ω |
| 2N5116, 2N5116UB | | | | | 175 | Ω |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Gate reverse current | 3411 | Bias condition C, $V_{GS}$ = 20 V dc; $V_{DS}$ = 0 | $I_{GSS}$ | | 1.0 | µA dc |
| Drain cutoff current | 3413 | Bias condition A; | $I_{D(off)2}$ | | | |
| 2N5114, 2N5114UB | | $V_{GS}$ = 12 V dc; $V_{DS}$ = -15 V dc | | | -1.0 | µA dc |
| 2N5115, 2N5115UB | | $V_{GS}$ = 7.0 V dc; $V_{DS}$ = -15 V dc | | | -1.0 | µA dc |
| 2N5116, 2N5116UB | | $V_{GS}$ = 5.0 V dc; $V_{DS}$ = -15 V dc | | | -1.0 | µA dc |
| Subgroup 4 | | | | | | |
| Small-signal common-source short-circuit input capacitance | 3431 | $V_{DS}$ = -15 V dc; $V_{GS}$ = 0 f = 1 MHz | $C_{iss}$ | | | |
| 2N5114, 2N5114UB | | | | | 25 | pF |
| 2N5115, 2N5115UB | | | | | 25 | pF |
| 2N5116, 2N5116UB | | | | | 27 | pF |
| Small-signal common-source short-circuit reverse transfer capacitance | 3433 | | $C_{rss}$ | | | |
| 2N5114, 2N5114UB | | $V_{DS}$ = 0; $V_{GS}$ = 12 V dc | | | | |
| 2N5115, 2N5115UB | | $V_{DS}$ = 0; $V_{GS}$ = 7.0 V dc | | | 7.0 | pF |
| 2N5116, 2N5116UB | | $V_{DS}$ = 0; $V_{GS}$ = 5.0 V dc | | | 7.0 | pF |
| | | | | | 7.0 | pF |
| Turn-on delay time | 3459 | See figure 3 | $t_{d(on)}$ | | | |
| 2N5114, 2N5114UB | | | | | 6 | ns |
| 2N5115, 2N5115UB | | | | | 10 | ns |
| 2N5116, 2N5116UB | | | | | 25 | ns |
| Rise time | 3459 | See figure 3 | $t_r$ | | | |
| 2N5114, 2N5114UB | | | | | 10 | ns |
| 2N5115, 2N5115UB | | | | | 20 | ns |
| 2N5116, 2N5116UB | | | | | 35 | ns |
| Turn-off delay time | 3459 | See figure 3 | $t_{d(off)}$ | | | |
| 2N5114, 2N5114UB | | | | | 6 | ns |
| 2N5115, 2N5115UB | | | | | 8 | ns |
| 2N5116, 2N5116UB | | | | | 20 | ns |

See footnote at end of table



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | $t_f$ | | | |
| Fall time | 3459 | See figure 3 | | | | |
| 2N5114, 2N5114UB | | | | | 15 | ns |
| 2N5115, 2N5115UB | | | | | 30 | ns |
| 2N5116, 2N5116UB | | | | | 60 | ns |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.

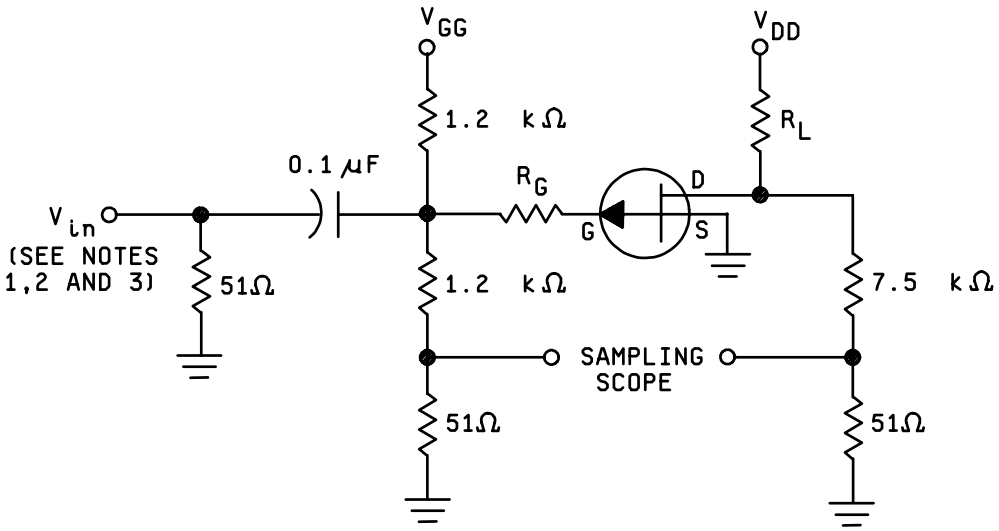
* TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 12 devices c = 0 |
| Temperature cycling | 1051 | -55°C to +150°C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak | | Test conditions G or H | |
| Gross leak | | Test conditions C or D | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 1/ | | | 12 devices c = 0 |
| Blocking life | 1048 | 1,000 hours minimum, $T_A$ = +150°C, $V_{DG}$ or $V_{GS}$ = 80 percent of rated. | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroups 4 and 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| ESD | 1020 | | |

1/ A separate sample for each test shall be pulled.



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TEST CIRCUIT

| Test conditions and component value | | | | | |
|---|---|---|---|---|---|
| Type | $V_{DD}$ | $V_{GG}$ | $R_L$ | $R_G$ | $I_{D(ON)}$ (1) |
| | V dc | V dc | ohms | ohms | μA dc |
| 2N5114, 2N5114UB | -10 | 20 | 430 | 100 | -15 |
| 2N5115, 2N5115UB | -6.0 | 12 | 910 | 220 | -7.0 |
| 2N5116, 2N5116UB | -6.0 | 8.0 | 2,000 | 390 | -3.0 |

(1) Nominal value; exact value varies slightly with transistor parameters.

NOTES:
1. The input waveform has the following characteristics: tp = 100 ns; $t_r \leq 1$ ns; $t_f \leq 1$ ns; duty cycle ≈ 2 percent. It is supplied by a generator with $Z_{out}$ = 50 ohms.
2. Resistor tolerance is ± 10 percent.
3. Voltage limits for the $V_{in}$ voltage (input switching levels) are as follows:
   2N5114, 2N5114UB $V_{in}$. = 11 V dc (from 0 v to -11 V dc signal level).
   2N5115, 2N5115UB $V_{in}$. = 7 V dc (from 0 v to -7 V dc signal level).
   2N5116, 2N5116UB $V_{in}$. = 5 V dc (from 0 v to -5 V dc signal level).

FIGURE 3. Switching time test circuit.



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AV, MI
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2006-024)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .



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